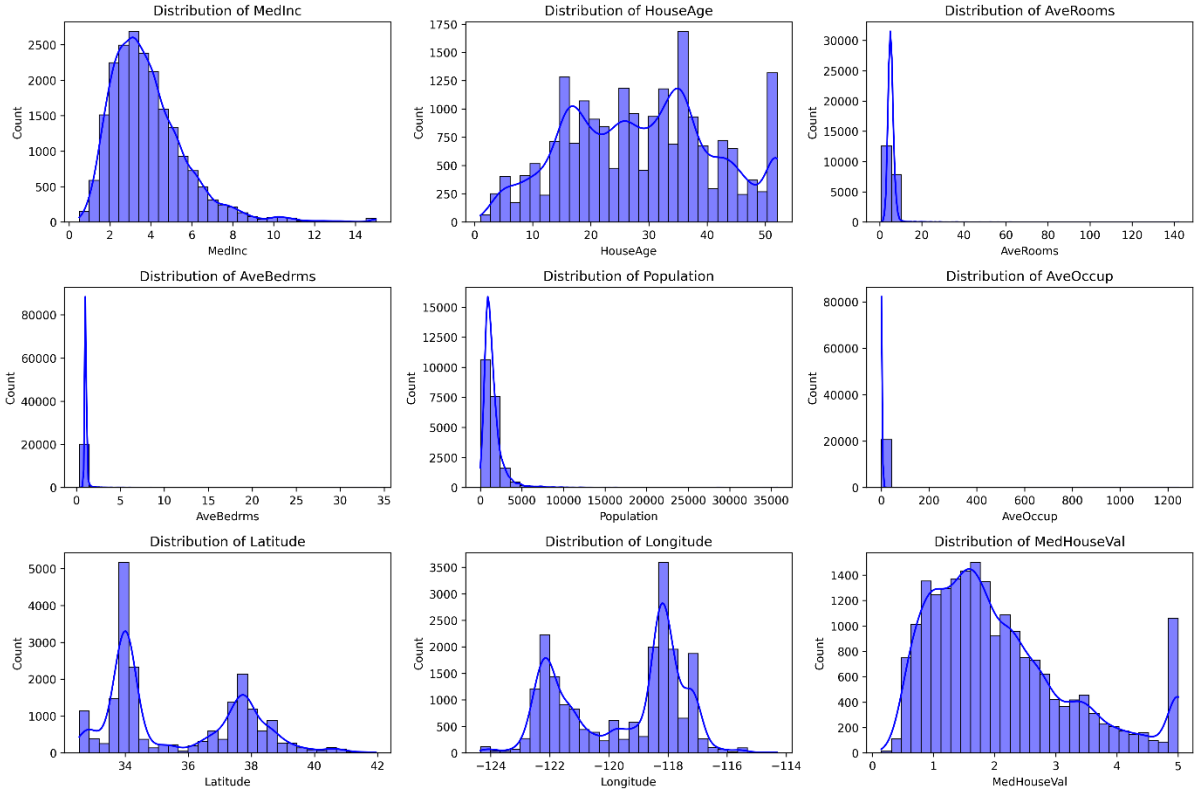
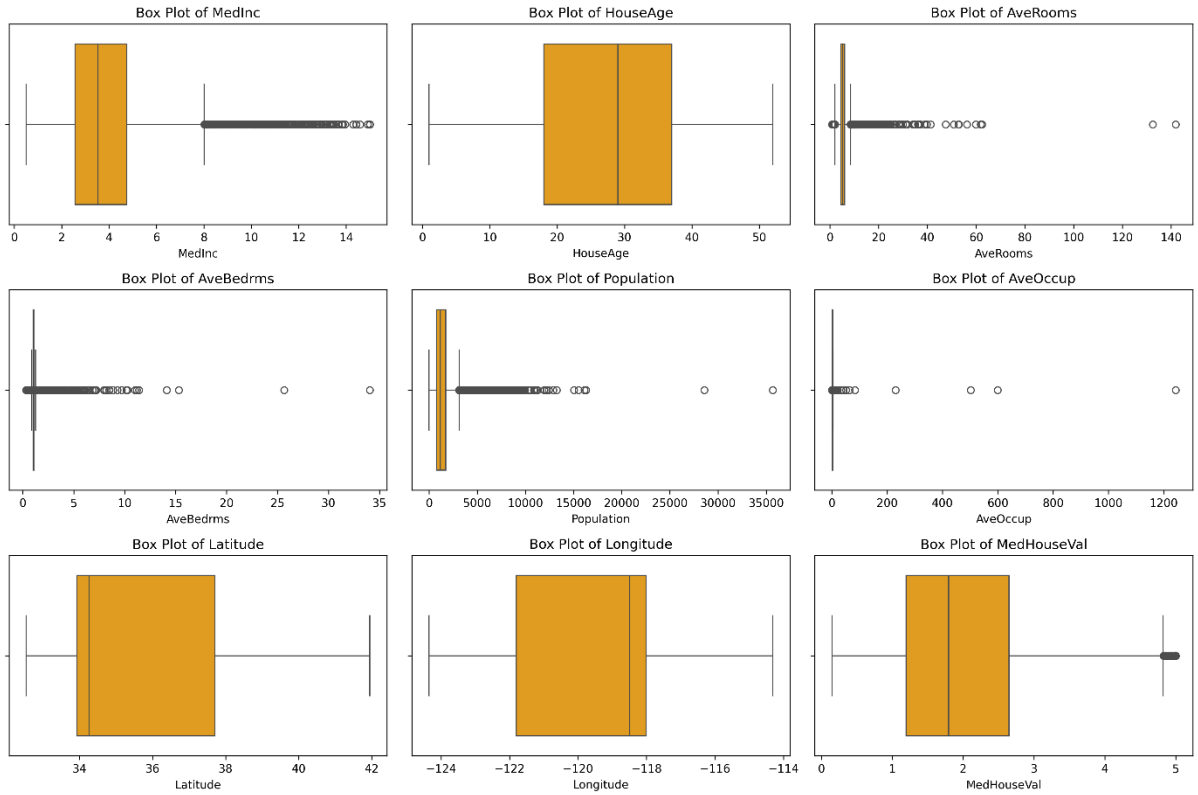
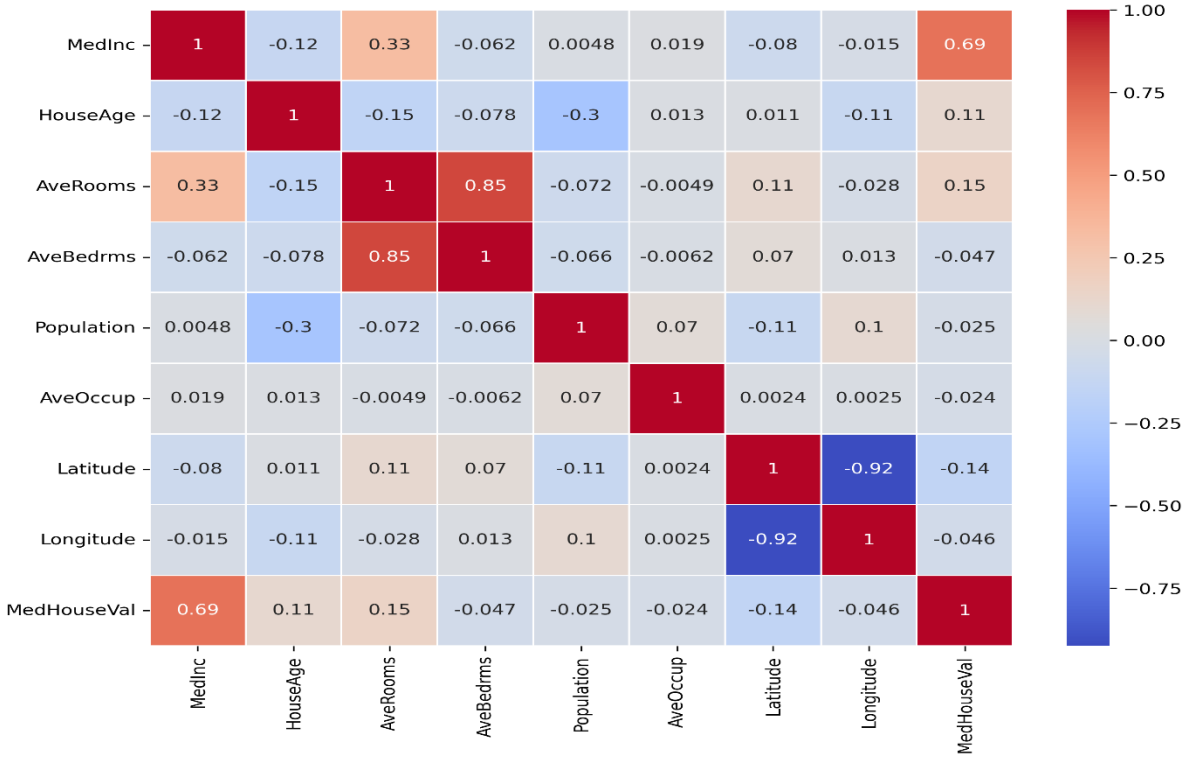
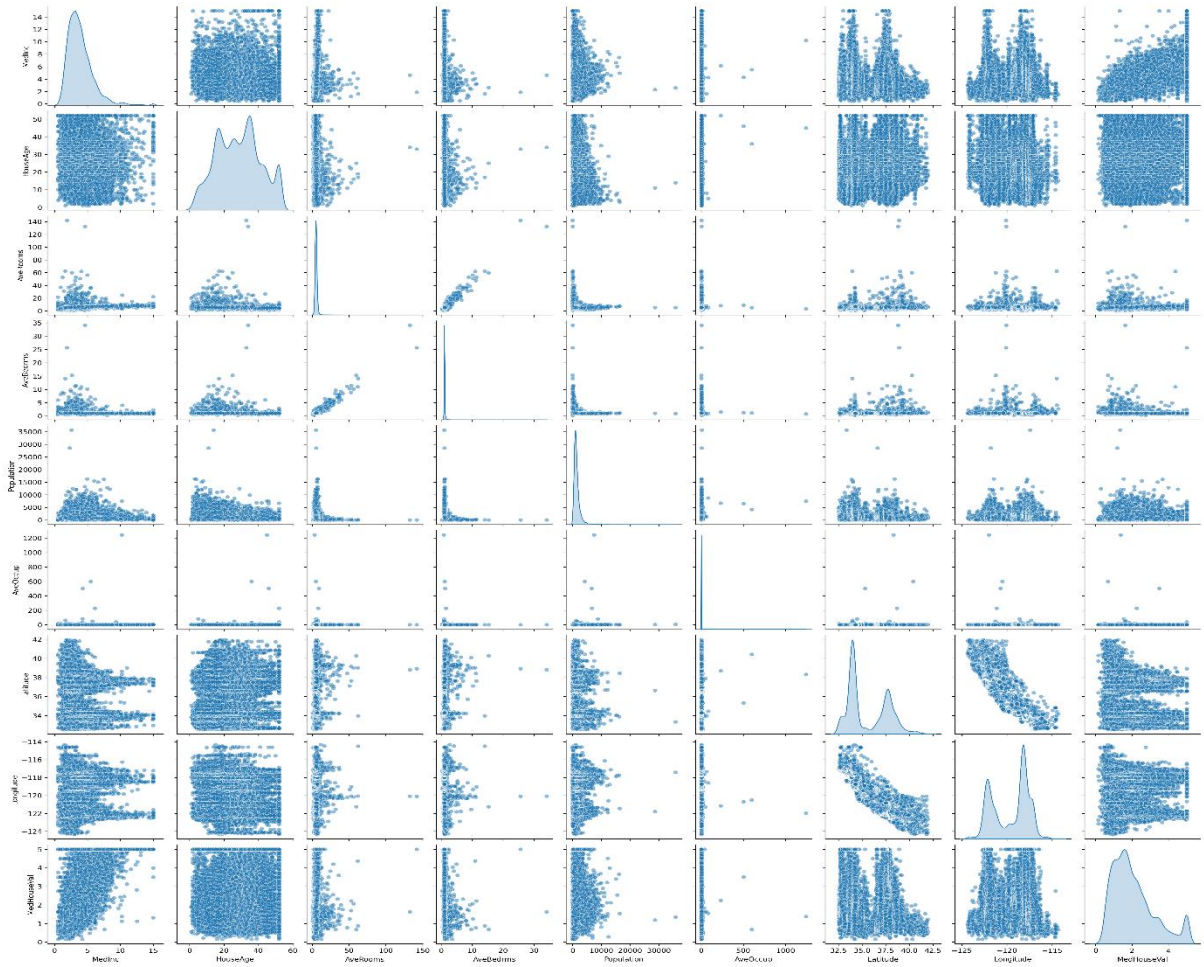
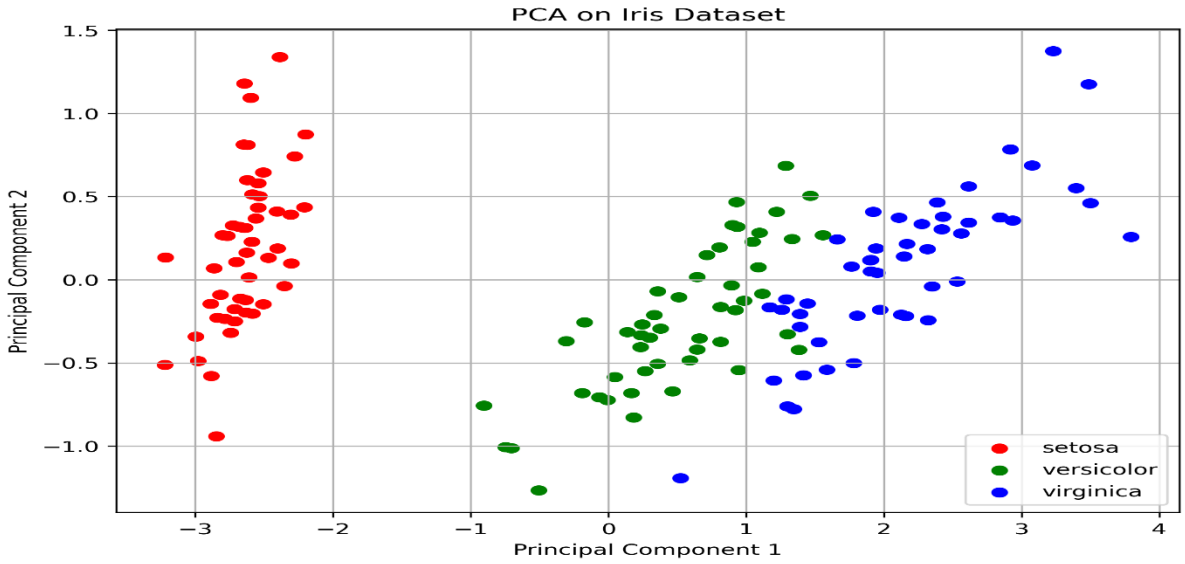
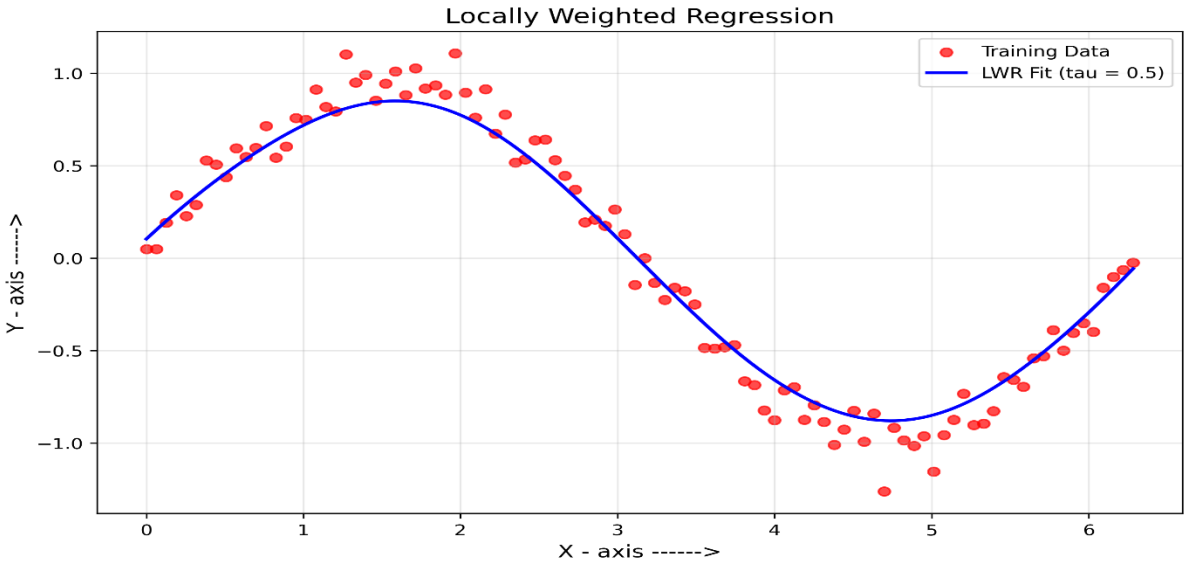

Distribution of MedInc
Count
MedInc
Distribution of HouseAge
Count
HouseAge
Distribution of AveRooms
Count
AveRooms
Distribution of AveBedrms
Count
AveBedrms
Distribution of Population
Count
Population
Distribution of AveOccup
Count
AveOccup
Distribution of Latitude
Count
Latitude
Distribution of Longitude
Count
Longitude
Distribution of MedHouseVal
Count
MedHouseVal

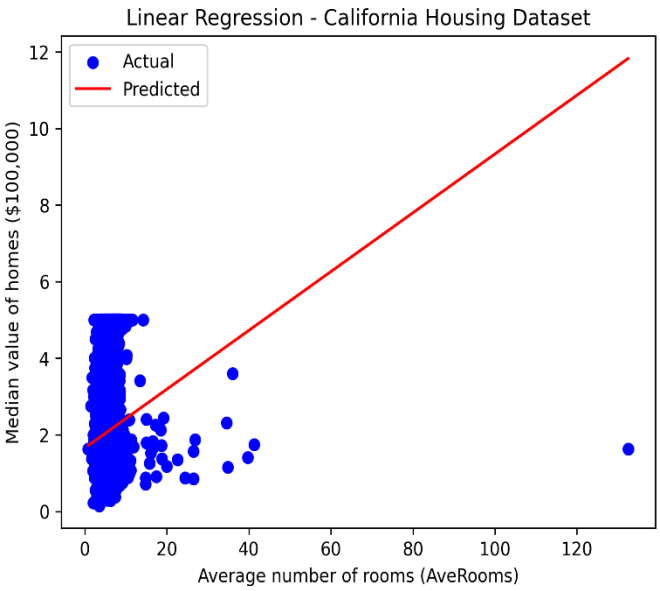

Box Plot of MedInc
MedInc
Box Plot of HouseAge
HouseAge
Box Plot of AveRooms
AveRooms
Box Plot of AveBedrms
AveBedrms
Box Plot of Population
Population
Box Plot of AveOccup
AveOccup
Box Plot of Latitude
Latitude
Box Plot of Longitude
Longitude
Box Plot of MedHouseVal
MedHouseVal

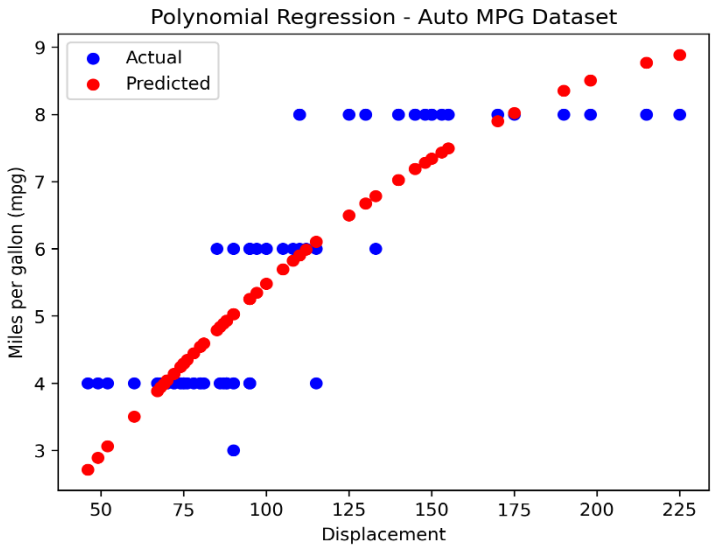

MedInc
HouseAge
AveRooms
AveBedrms
Population
AveOccup
Latitude
Longitude
MedHouseVal
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

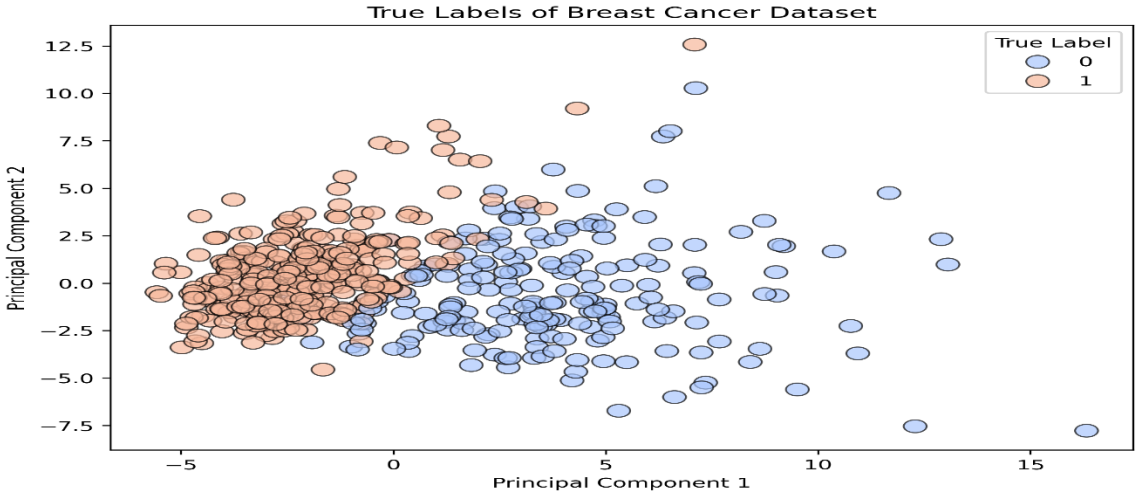

MedInc
HouseAge
AveRooms
AveBedrms
Population
AveOccup
Latitude
Longitude
MedHouseVal

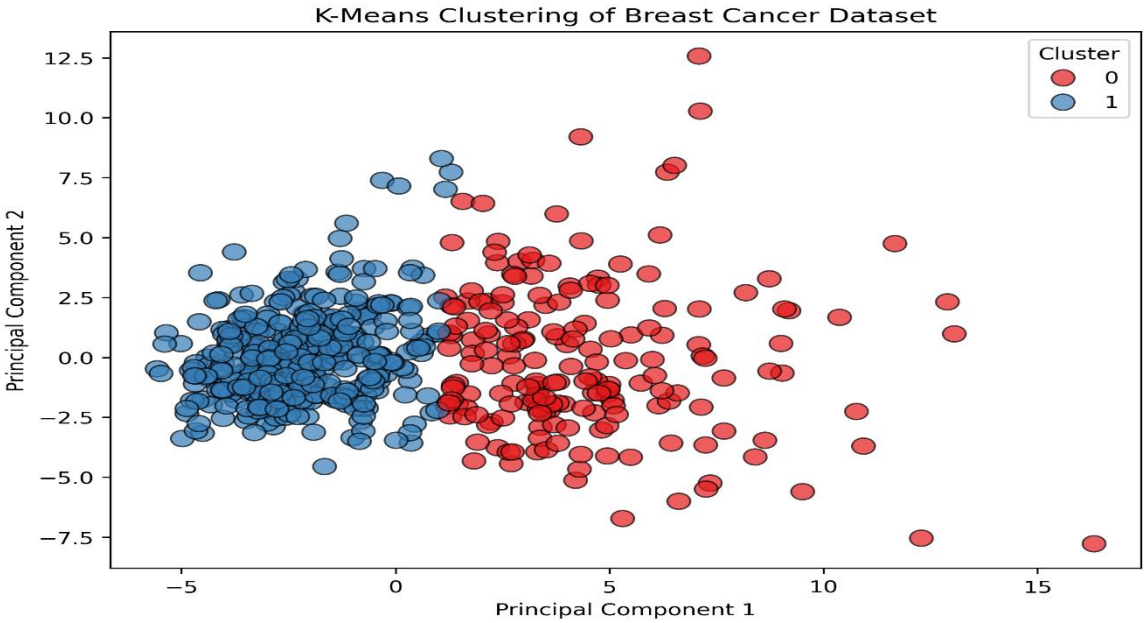

PCA on Iris Dataset
Principal Component 2
Principal Component 1
setosa
versicolor
virginica

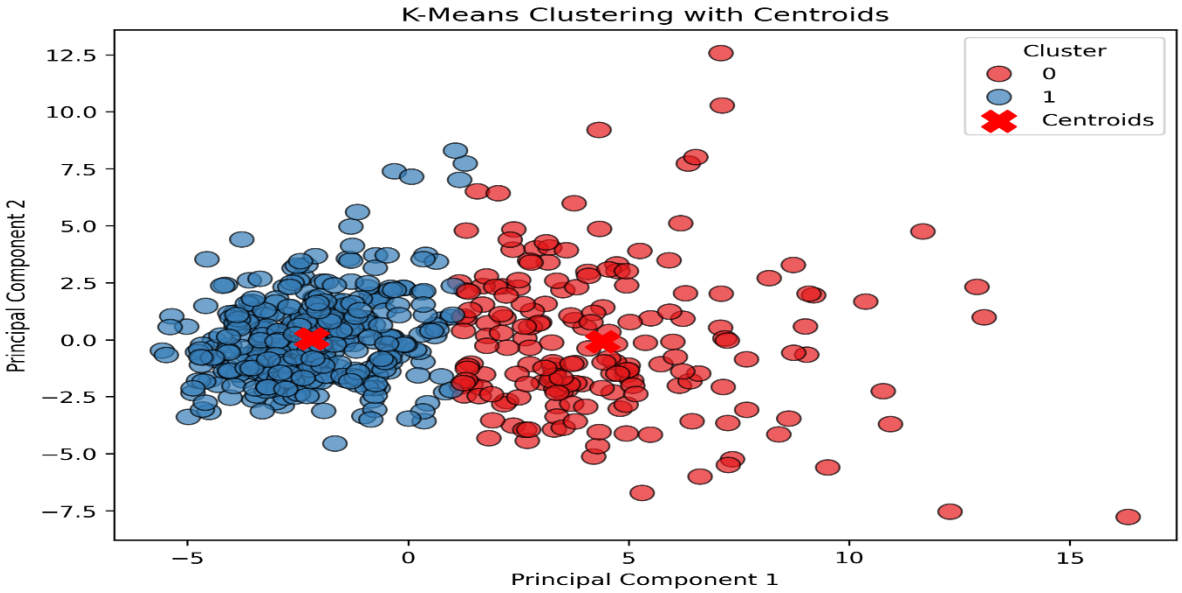

Locally Weighted Regression
Training Data
LWR Fit (tau = 0.5)
Y - axis ------>
X - axis ------>


Linear Regression - California Housing Dataset
Actual
Predicted
Median value of homes ($100,000)
Average number of rooms (AveRooms)


Polynomial Regression - Auto MPG Dataset
Actual
Predicted
Miles per gallon (mpg)
Displacement


True Labels of Breast Cancer Dataset
True Label
0
1
Principal Component 2
Principal Component 1
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−5
0
5
10
15


K-Means Clustering of Breast Cancer Dataset
Cluster
0
1
Principal Component 2
Principal Component 1
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−5
0
5
10
15


K-Means Clustering with Centroids
Cluster
0
1
Centroids
Principal Component 2
Principal Component 1
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−5
0
5
10
15

```
Results for k = 5:
Point x 51 (value :  0.2111) is classified as Class1
Point x 52 (value :  0.5023) is classified as Class2
Point x 53 (value :  0.0836) is classified as Class1
Point x 54 (value :  0.3225) is classified as Class1
Point x 55 (value :  0.2640) is classified as Class1
Point x 56 (value :  0.4189) is classified as Class1
Point x 57 (value :  0.0180) is classified as Class1
Point x 58 (value :  0.9803) is classified as Class2
Point x 59 (value :  0.8247) is classified as Class2
Point x 60 (value :  0.3510) is classified as Class1
Point x 61 (value :  0.3824) is classified as Class1
Point x 62 (value :  0.5808) is classified as Class2
Point x 63 (value :  0.9564) is classified as Class2
Point x 64 (value :  0.3450) is classified as Class1
Point x 65 (value :  0.1428) is classified as Class1
Point x 66 (value :  0.0906) is classified as Class1
Point x 67 (value :  0.1508) is classified as Class1
Point x 68 (value :  0.1939) is classified as Class1
Point x 69 (value :  0.3222) is classified as Class1
Point x 70 (value :  0.7759) is classified as Class2
Point x 71 (value :  0.8437) is classified as Class2
Point x 72 (value :  0.9427) is classified as Class2
Point x 73 (value :  0.2453) is classified as Class1
Point x 74 (value :  0.2588) is classified as Class1
Point x 75 (value :  0.8446) is classified as Class2
Point x 76 (value :  0.6839) is classified as Class2
Point x 77 (value :  0.5486) is classified as Class2
Point x 78 (value :  0.9124) is classified as Class2
Point x 79 (value :  0.5082) is classified as Class2
Point x 80 (value :  0.8065) is classified as Class2
Point x 81 (value :  0.4547) is classified as Class1
Point x 82 (value :  0.4683) is classified as Class1
Point x 83 (value :  0.5625) is classified as Class2
Point x 84 (value :  0.4457) is classified as Class1
Point x 85 (value :  0.2863) is classified as Class1
Point x 86 (value :  0.8984) is classified as Class2
Point x 87 (value :  0.6882) is classified as Class2
Point x 88 (value :  0.7494) is classified as Class2
Point x 89 (value :  0.3507) is classified as Class1
Point x 90 (value :  0.5408) is classified as Class2
Point x 91 (value :  0.6724) is classified as Class2
Point x 92 (value :  0.0025) is classified as Class1
Point x 93 (value :  0.0643) is classified as Class1
Point x 94 (value :  0.6097) is classified as Class2
Point x 95 (value :  0.5430) is classified as Class2
Point x 96 (value :  0.5959) is classified as Class2
Point x 97 (value :  0.3600) is classified as Class1
Point x 98 (value :  0.1831) is classified as Class1
Point x 99 (value :  0.3428) is classified as Class1
Point x 100 (value :  0.5131) is classified as Class2
```

```
Results for k = 20:
Point x 51 (value :  0.2111) is classified as Class1
Point x 52 (value :  0.5023) is classified as Class2
Point x 53 (value :  0.0836) is classified as Class1
Point x 54 (value :  0.3225) is classified as Class1
Point x 55 (value :  0.2640) is classified as Class1
Point x 56 (value :  0.4189) is classified as Class1
Point x 57 (value :  0.0180) is classified as Class1
Point x 58 (value :  0.9803) is classified as Class2
Point x 59 (value :  0.8247) is classified as Class2
Point x 60 (value :  0.3510) is classified as Class1
Point x 61 (value :  0.3824) is classified as Class1
Point x 62 (value :  0.5808) is classified as Class2
Point x 63 (value :  0.9564) is classified as Class2
Point x 64 (value :  0.3450) is classified as Class1
Point x 65 (value :  0.1428) is classified as Class1
Point x 66 (value :  0.0906) is classified as Class1
Point x 67 (value :  0.1508) is classified as Class1
Point x 68 (value :  0.1939) is classified as Class1
Point x 69 (value :  0.3222) is classified as Class1
Point x 70 (value :  0.7759) is classified as Class2
Point x 71 (value :  0.8437) is classified as Class2
Point x 72 (value :  0.9427) is classified as Class2
Point x 73 (value :  0.2453) is classified as Class1
Point x 74 (value :  0.2588) is classified as Class1
Point x 75 (value :  0.8446) is classified as Class2
Point x 76 (value :  0.6839) is classified as Class2
Point x 77 (value :  0.5486) is classified as Class2
Point x 78 (value :  0.9124) is classified as Class2
Point x 79 (value :  0.5082) is classified as Class2
Point x 80 (value :  0.8065) is classified as Class2
Point x 81 (value :  0.4547) is classified as Class1
Point x 82 (value :  0.4683) is classified as Class1
Point x 83 (value :  0.5625) is classified as Class2
Point x 84 (value :  0.4457) is classified as Class1
Point x 85 (value :  0.2863) is classified as Class1
Point x 86 (value :  0.8984) is classified as Class2
Point x 87 (value :  0.6882) is classified as Class2
Point x 88 (value :  0.7494) is classified as Class2
Point x 89 (value :  0.3507) is classified as Class1
Point x 90 (value :  0.5408) is classified as Class2
Point x 91 (value :  0.6724) is classified as Class2
Point x 92 (value :  0.0025) is classified as Class1
Point x 93 (value :  0.0643) is classified as Class1
Point x 94 (value :  0.6097) is classified as Class2
Point x 95 (value :  0.5430) is classified as Class2
Point x 96 (value :  0.5959) is classified as Class2
Point x 97 (value :  0.3600) is classified as Class1
Point x 98 (value :  0.1831) is classified as Class1
Point x 99 (value :  0.3428) is classified as Class1
Point x 100 (value :  0.5131) is classified as Class2
```

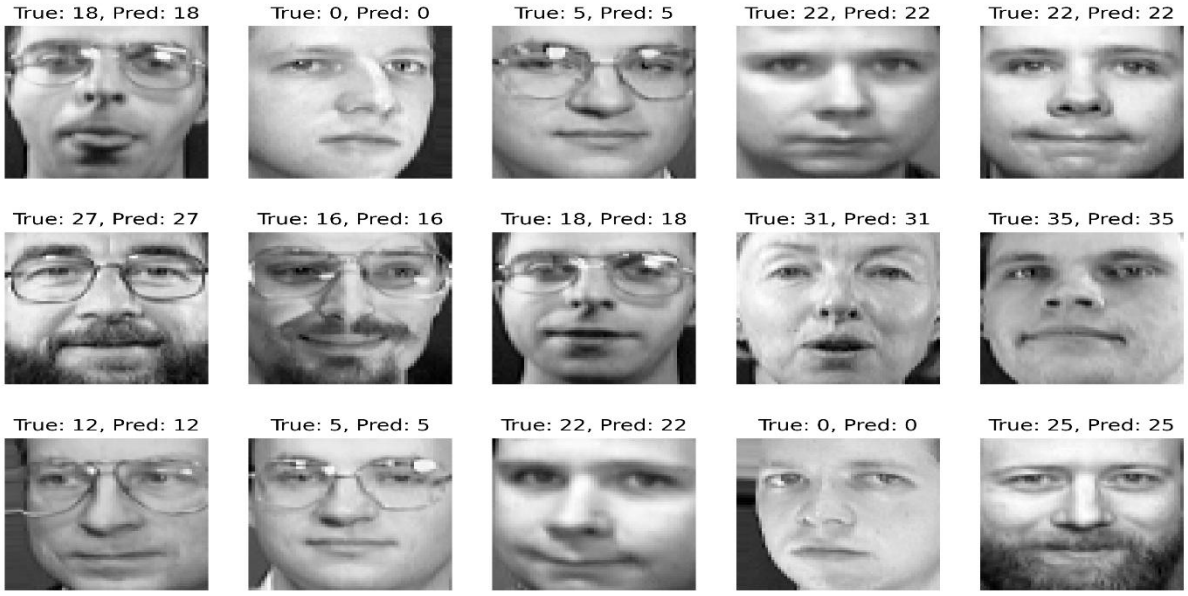
```
Results for k = 30:
Point x 51 (value :  0.2111) is classified as Class1
Point x 52 (value :  0.5023) is classified as Class2
Point x 53 (value :  0.0836) is classified as Class1
Point x 54 (value :  0.3225) is classified as Class1
Point x 55 (value :  0.2640) is classified as Class1
Point x 56 (value :  0.4189) is classified as Class2
Point x 57 (value :  0.0180) is classified as Class1
Point x 58 (value :  0.9803) is classified as Class2
Point x 59 (value :  0.8247) is classified as Class2
Point x 60 (value :  0.3510) is classified as Class1
Point x 61 (value :  0.3824) is classified as Class1
Point x 62 (value :  0.5808) is classified as Class2
Point x 63 (value :  0.9564) is classified as Class2
Point x 64 (value :  0.3450) is classified as Class1
Point x 65 (value :  0.1428) is classified as Class1
Point x 66 (value :  0.0906) is classified as Class1
Point x 67 (value :  0.1508) is classified as Class1
Point x 68 (value :  0.1939) is classified as Class1
Point x 69 (value :  0.3222) is classified as Class1
Point x 70 (value :  0.7759) is classified as Class2
Point x 71 (value :  0.8437) is classified as Class2
Point x 72 (value :  0.9427) is classified as Class2
Point x 73 (value :  0.2453) is classified as Class1
Point x 74 (value :  0.2588) is classified as Class1
Point x 75 (value :  0.8446) is classified as Class2
Point x 76 (value :  0.6839) is classified as Class2
Point x 77 (value :  0.5486) is classified as Class2
Point x 78 (value :  0.9124) is classified as Class2
Point x 79 (value :  0.5082) is classified as Class2
Point x 80 (value :  0.8065) is classified as Class2
Point x 81 (value :  0.4547) is classified as Class2
Point x 82 (value :  0.4683) is classified as Class2
Point x 83 (value :  0.5625) is classified as Class2
Point x 84 (value :  0.4457) is classified as Class2
Point x 85 (value :  0.2863) is classified as Class1
Point x 86 (value :  0.8984) is classified as Class2
Point x 87 (value :  0.6882) is classified as Class2
Point x 88 (value :  0.7494) is classified as Class2
Point x 89 (value :  0.3507) is classified as Class1
Point x 90 (value :  0.5408) is classified as Class2
Point x 91 (value :  0.6724) is classified as Class2
Point x 92 (value :  0.0025) is classified as Class1
Point x 93 (value :  0.0643) is classified as Class1
Point x 94 (value :  0.6097) is classified as Class2
Point x 95 (value :  0.5430) is classified as Class2
Point x 96 (value :  0.5959) is classified as Class2
Point x 97 (value :  0.3600) is classified as Class1
Point x 98 (value :  0.1831) is classified as Class1
Point x 99 (value :  0.3428) is classified as Class1
Point x 100 (value :  0.5131) is classified as Class2

Classification complete.
```

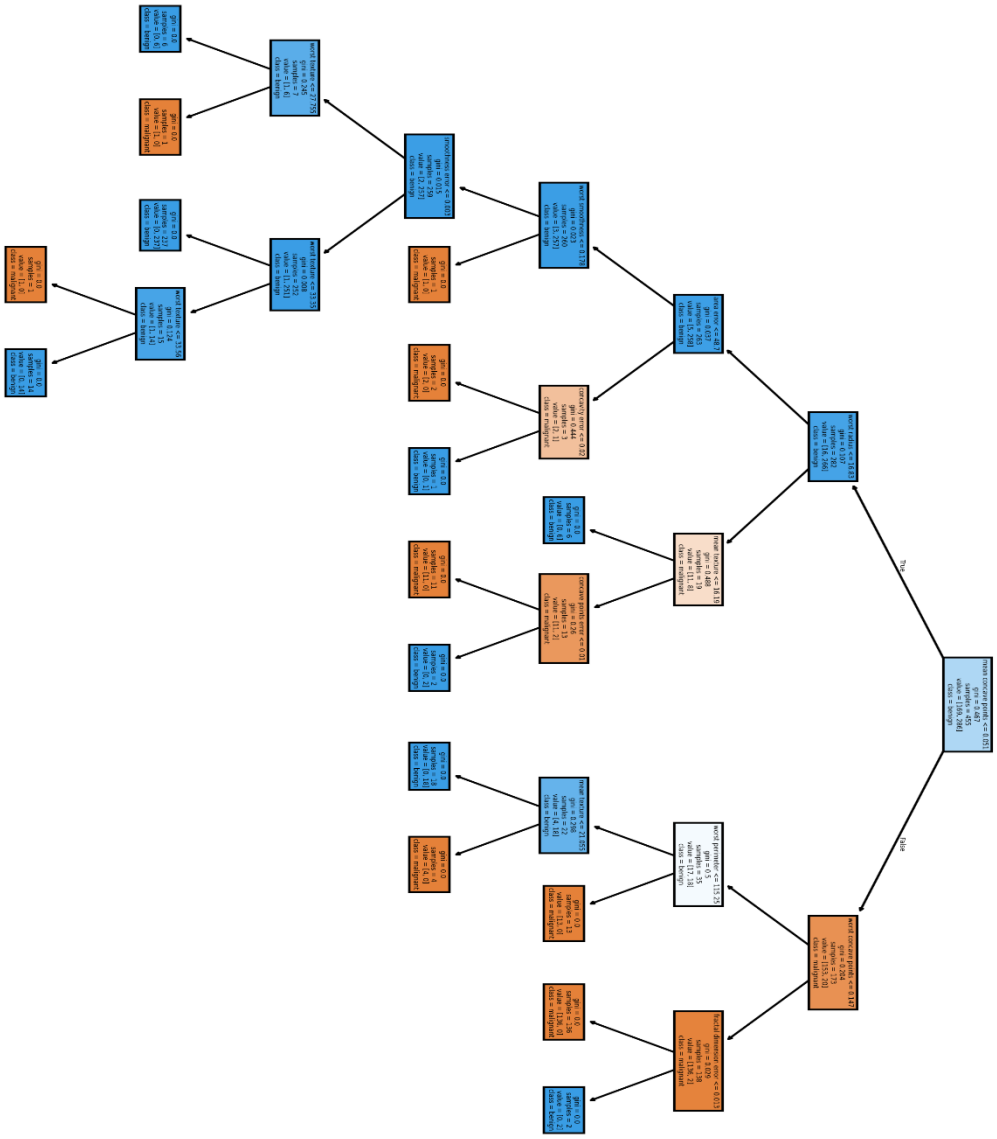
```
Results for k = 1:
Point x 51 (value :  0.2111) is classified as Class1
Point x 52 (value :  0.5023) is classified as Class1
Point x 53 (value :  0.0836) is classified as Class1
Point x 54 (value :  0.3225) is classified as Class1
Point x 55 (value :  0.2640) is classified as Class1
Point x 56 (value :  0.4189) is classified as Class1
Point x 57 (value :  0.0180) is classified as Class1
Point x 58 (value :  0.9803) is classified as Class2
Point x 59 (value :  0.8247) is classified as Class2
Point x 60 (value :  0.3510) is classified as Class1
Point x 61 (value :  0.3824) is classified as Class1
Point x 62 (value :  0.5808) is classified as Class2
Point x 63 (value :  0.9564) is classified as Class2
Point x 64 (value :  0.3450) is classified as Class1
Point x 65 (value :  0.1428) is classified as Class1
Point x 66 (value :  0.0906) is classified as Class1
Point x 67 (value :  0.1508) is classified as Class1
Point x 68 (value :  0.1939) is classified as Class1
Point x 69 (value :  0.3222) is classified as Class1
Point x 70 (value :  0.7759) is classified as Class2
Point x 71 (value :  0.8437) is classified as Class2
Point x 72 (value :  0.9427) is classified as Class2
Point x 73 (value :  0.2453) is classified as Class1
Point x 74 (value :  0.2588) is classified as Class1
Point x 75 (value :  0.8446) is classified as Class2
Point x 76 (value :  0.6839) is classified as Class2
Point x 77 (value :  0.5486) is classified as Class2
Point x 78 (value :  0.9124) is classified as Class2
Point x 79 (value :  0.5082) is classified as Class2
Point x 80 (value :  0.8065) is classified as Class2
Point x 81 (value :  0.4547) is classified as Class1
Point x 82 (value :  0.4683) is classified as Class1
Point x 83 (value :  0.5625) is classified as Class2
Point x 84 (value :  0.4457) is classified as Class1
Point x 85 (value :  0.2863) is classified as Class1
Point x 86 (value :  0.8984) is classified as Class2
Point x 87 (value :  0.6882) is classified as Class2
Point x 88 (value :  0.7494) is classified as Class2
Point x 89 (value :  0.3507) is classified as Class1
Point x 90 (value :  0.5408) is classified as Class2
Point x 91 (value :  0.6724) is classified as Class2
Point x 92 (value :  0.0025) is classified as Class1
Point x 93 (value :  0.0643) is classified as Class1
Point x 94 (value :  0.6097) is classified as Class2
Point x 95 (value :  0.5430) is classified as Class2
Point x 96 (value :  0.5959) is classified as Class2
Point x 97 (value :  0.3600) is classified as Class1
Point x 98 (value :  0.1831) is classified as Class1
Point x 99 (value :  0.3428) is classified as Class1
Point x 100 (value :  0.5131) is classified as Class2
```

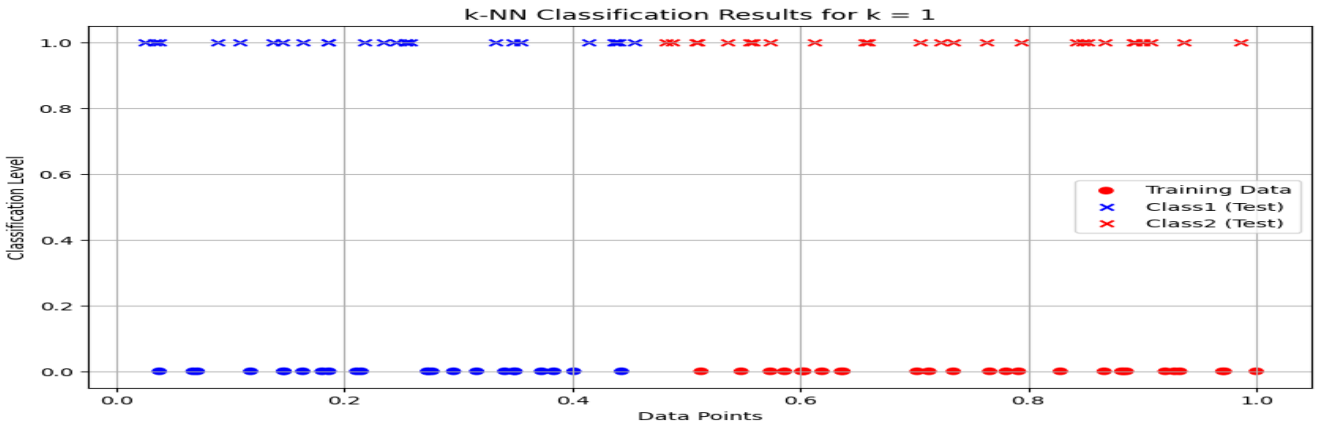
```
Results for k = 2:
Point x 51 (value :  0.2111) is classified as Class1
Point x 52 (value :  0.5023) is classified as Class1
Point x 53 (value :  0.0836) is classified as Class1
Point x 54 (value :  0.3225) is classified as Class1
Point x 55 (value :  0.2640) is classified as Class1
Point x 56 (value :  0.4189) is classified as Class1
Point x 57 (value :  0.0180) is classified as Class1
Point x 58 (value :  0.9803) is classified as Class2
Point x 59 (value :  0.8247) is classified as Class2
Point x 60 (value :  0.3510) is classified as Class1
Point x 61 (value :  0.3824) is classified as Class1
Point x 62 (value :  0.5808) is classified as Class2
Point x 63 (value :  0.9564) is classified as Class2
Point x 64 (value :  0.3450) is classified as Class1
Point x 65 (value :  0.1428) is classified as Class1
Point x 66 (value :  0.0906) is classified as Class1
Point x 67 (value :  0.1508) is classified as Class1
Point x 68 (value :  0.1939) is classified as Class1
Point x 69 (value :  0.3222) is classified as Class1
Point x 70 (value :  0.7759) is classified as Class2
Point x 71 (value :  0.8437) is classified as Class2
Point x 72 (value :  0.9427) is classified as Class2
Point x 73 (value :  0.2453) is classified as Class1
Point x 74 (value :  0.2588) is classified as Class1
Point x 75 (value :  0.8446) is classified as Class2
Point x 76 (value :  0.6839) is classified as Class2
Point x 77 (value :  0.5486) is classified as Class2
Point x 78 (value :  0.9124) is classified as Class2
Point x 79 (value :  0.5082) is classified as Class2
Point x 80 (value :  0.8065) is classified as Class2
Point x 81 (value :  0.4547) is classified as Class1
Point x 82 (value :  0.4683) is classified as Class1
Point x 83 (value :  0.5625) is classified as Class2
Point x 84 (value :  0.4457) is classified as Class1
Point x 85 (value :  0.2863) is classified as Class1
Point x 86 (value :  0.8984) is classified as Class2
Point x 87 (value :  0.6882) is classified as Class2
Point x 88 (value :  0.7494) is classified as Class2
Point x 89 (value :  0.3507) is classified as Class1
Point x 90 (value :  0.5408) is classified as Class2
Point x 91 (value :  0.6724) is classified as Class2
Point x 92 (value :  0.0025) is classified as Class1
Point x 93 (value :  0.0643) is classified as Class1
Point x 94 (value :  0.6097) is classified as Class2
Point x 95 (value :  0.5430) is classified as Class2
Point x 96 (value :  0.5959) is classified as Class2
Point x 97 (value :  0.3600) is classified as Class1
Point x 98 (value :  0.1831) is classified as Class1
Point x 99 (value :  0.3428) is classified as Class1
Point x 100 (value :  0.5131) is classified as Class2
```

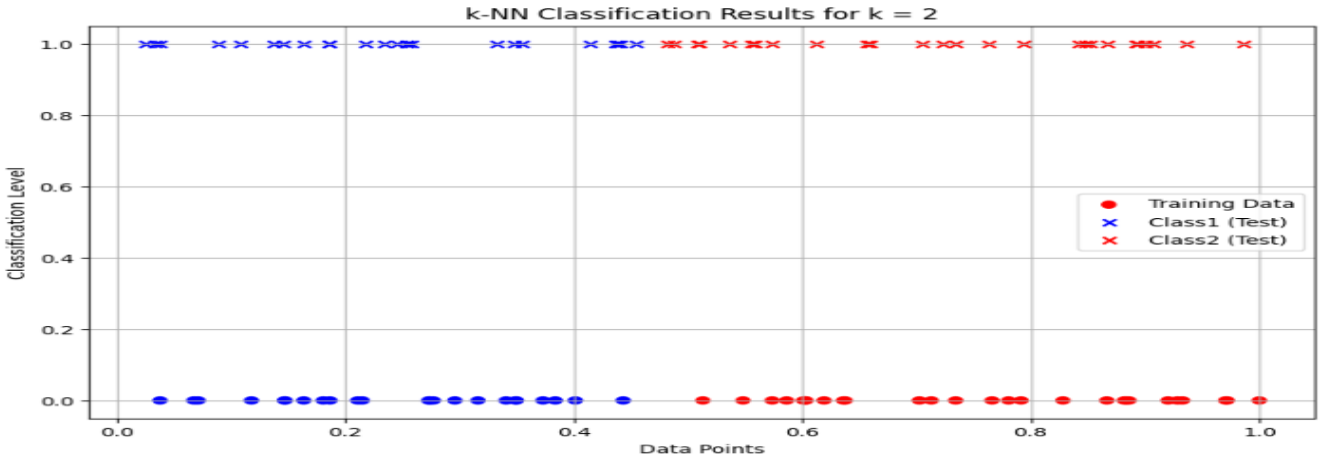
```
Results for k = 3:
Point x 51 (value :  0.2111) is classified as Class1
Point x 52 (value :  0.5023) is classified as Class2
Point x 53 (value :  0.0836) is classified as Class1
Point x 54 (value :  0.3225) is classified as Class1
Point x 55 (value :  0.2640) is classified as Class1
Point x 56 (value :  0.4189) is classified as Class1
Point x 57 (value :  0.0180) is classified as Class1
Point x 58 (value :  0.9803) is classified as Class2
Point x 59 (value :  0.8247) is classified as Class2
Point x 60 (value :  0.3510) is classified as Class1
Point x 61 (value :  0.3824) is classified as Class1
Point x 62 (value :  0.5808) is classified as Class2
Point x 63 (value :  0.9564) is classified as Class2
Point x 64 (value :  0.3450) is classified as Class1
Point x 65 (value :  0.1428) is classified as Class1
Point x 66 (value :  0.0906) is classified as Class1
Point x 67 (value :  0.1508) is classified as Class1
Point x 68 (value :  0.1939) is classified as Class1
Point x 69 (value :  0.3222) is classified as Class1
Point x 70 (value :  0.7759) is classified as Class2
Point x 71 (value :  0.8437) is classified as Class2
Point x 72 (value :  0.9427) is classified as Class2
Point x 73 (value :  0.2453) is classified as Class1
Point x 74 (value :  0.2588) is classified as Class1
Point x 75 (value :  0.8446) is classified as Class2
Point x 76 (value :  0.6839) is classified as Class2
Point x 77 (value :  0.5486) is classified as Class2
Point x 78 (value :  0.9124) is classified as Class2
Point x 79 (value :  0.5082) is classified as Class2
Point x 80 (value :  0.8065) is classified as Class2
Point x 81 (value :  0.4547) is classified as Class1
Point x 82 (value :  0.4683) is classified as Class1
Point x 83 (value :  0.5625) is classified as Class2
Point x 84 (value :  0.4457) is classified as Class1
Point x 85 (value :  0.2863) is classified as Class1
Point x 86 (value :  0.8984) is classified as Class2
Point x 87 (value :  0.6882) is classified as Class2
Point x 88 (value :  0.7494) is classified as Class2
Point x 89 (value :  0.3507) is classified as Class1
Point x 90 (value :  0.5408) is classified as Class2
Point x 91 (value :  0.6724) is classified as Class2
Point x 92 (value :  0.0025) is classified as Class1
Point x 93 (value :  0.0643) is classified as Class1
Point x 94 (value :  0.6097) is classified as Class2
Point x 95 (value :  0.5430) is classified as Class2
Point x 96 (value :  0.5959) is classified as Class2
Point x 97 (value :  0.3600) is classified as Class1
Point x 98 (value :  0.1831) is classified as Class1
Point x 99 (value :  0.3428) is classified as Class1
Point x 100 (value :  0.5131) is classified as Class2
```

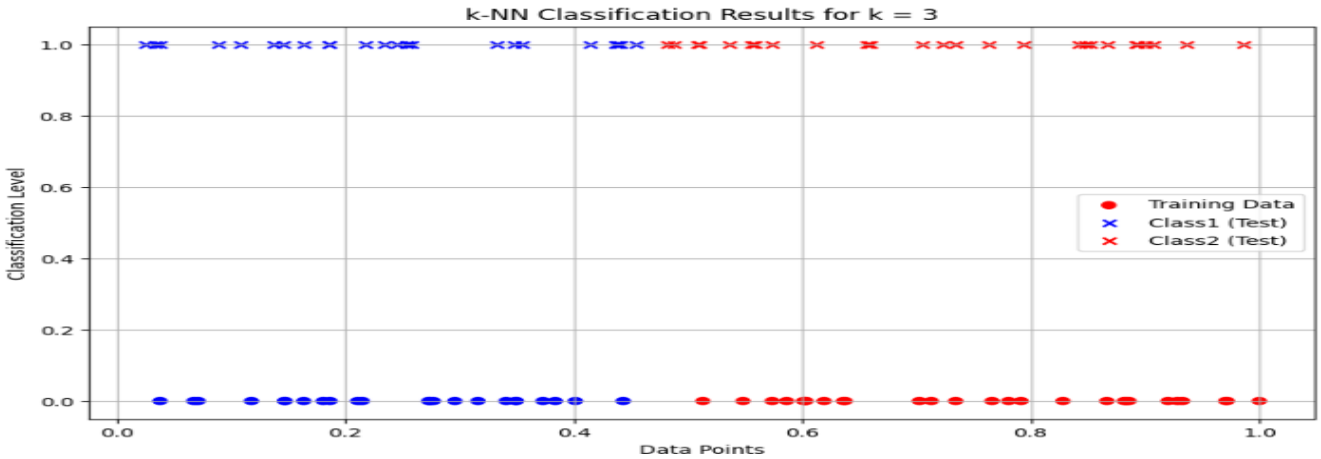
```
Results for k = 4:
Point x 51 (value :  0.2111) is classified as Class1
Point x 52 (value :  0.5023) is classified as Class2
Point x 53 (value :  0.0836) is classified as Class1
Point x 54 (value :  0.3225) is classified as Class1
Point x 55 (value :  0.2640) is classified as Class1
Point x 56 (value :  0.4189) is classified as Class1
Point x 57 (value :  0.0180) is classified as Class1
Point x 58 (value :  0.9803) is classified as Class2
Point x 59 (value :  0.8247) is classified as Class2
Point x 60 (value :  0.3510) is classified as Class1
Point x 61 (value :  0.3824) is classified as Class1
Point x 62 (value :  0.5808) is classified as Class2
Point x 63 (value :  0.9564) is classified as Class2
Point x 64 (value :  0.3450) is classified as Class1
Point x 65 (value :  0.1428) is classified as Class1
Point x 66 (value :  0.0906) is classified as Class1
Point x 67 (value :  0.1508) is classified as Class1
Point x 68 (value :  0.1939) is classified as Class1
Point x 69 (value :  0.3222) is classified as Class1
Point x 70 (value :  0.7759) is classified as Class2
Point x 71 (value :  0.8437) is classified as Class2
Point x 72 (value :  0.9427) is classified as Class2
Point x 73 (value :  0.2453) is classified as Class1
Point x 74 (value :  0.2588) is classified as Class1
Point x 75 (value :  0.8446) is classified as Class2
Point x 76 (value :  0.6839) is classified as Class2
Point x 77 (value :  0.5486) is classified as Class2
Point x 78 (value :  0.9124) is classified as Class2
Point x 79 (value :  0.5082) is classified as Class2
Point x 80 (value :  0.8065) is classified as Class2
Point x 81 (value :  0.4547) is classified as Class1
Point x 82 (value :  0.4683) is classified as Class1
Point x 83 (value :  0.5625) is classified as Class2
Point x 84 (value :  0.4457) is classified as Class1
Point x 85 (value :  0.2863) is classified as Class1
Point x 86 (value :  0.8984) is classified as Class2
Point x 87 (value :  0.6882) is classified as Class2
Point x 88 (value :  0.7494) is classified as Class2
Point x 89 (value :  0.3507) is classified as Class1
Point x 90 (value :  0.5408) is classified as Class2
Point x 91 (value :  0.6724) is classified as Class2
Point x 92 (value :  0.0025) is classified as Class1
Point x 93 (value :  0.0643) is classified as Class1
Point x 94 (value :  0.6097) is classified as Class2
Point x 95 (value :  0.5430) is classified as Class2
Point x 96 (value :  0.5959) is classified as Class2
Point x 97 (value :  0.3600) is classified as Class1
Point x 98 (value :  0.1831) is classified as Class1
Point x 99 (value :  0.3428) is classified as Class1
Point x 100 (value :  0.5131) is classified as Class2
```

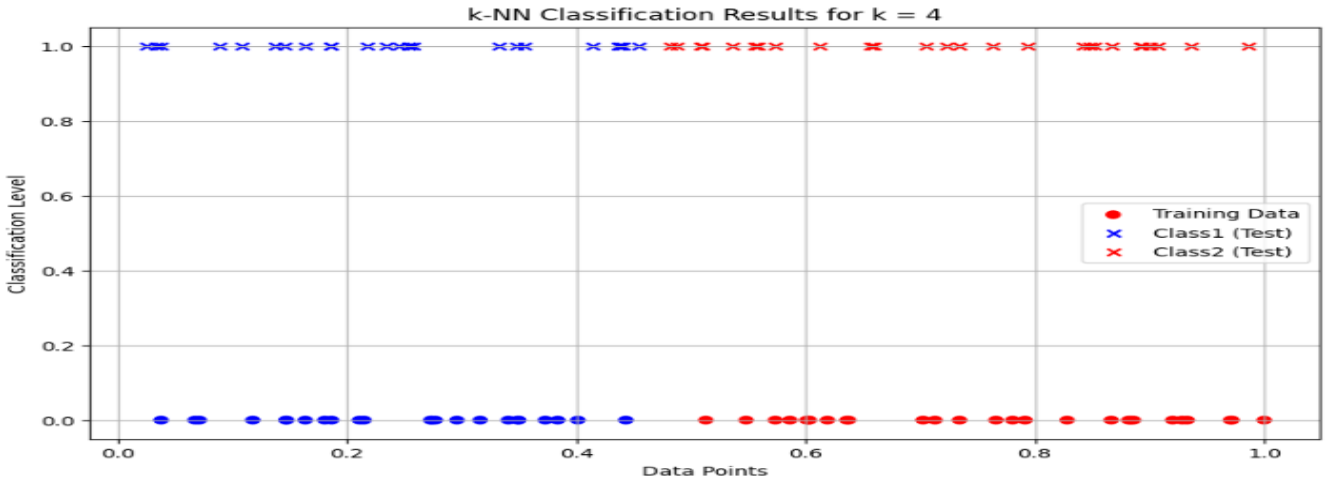

True: 18, Pred: 18
True: 0, Pred: 0
True: 5, Pred: 5
True: 22, Pred: 22
True: 22, Pred: 22
True: 27, Pred: 27
True: 16, Pred: 16
True: 18, Pred: 18
True: 31, Pred: 31
True: 35, Pred: 35
True: 12, Pred: 12
True: 5, Pred: 5
True: 22, Pred: 22
True: 0, Pred: 0
True: 25, Pred: 25

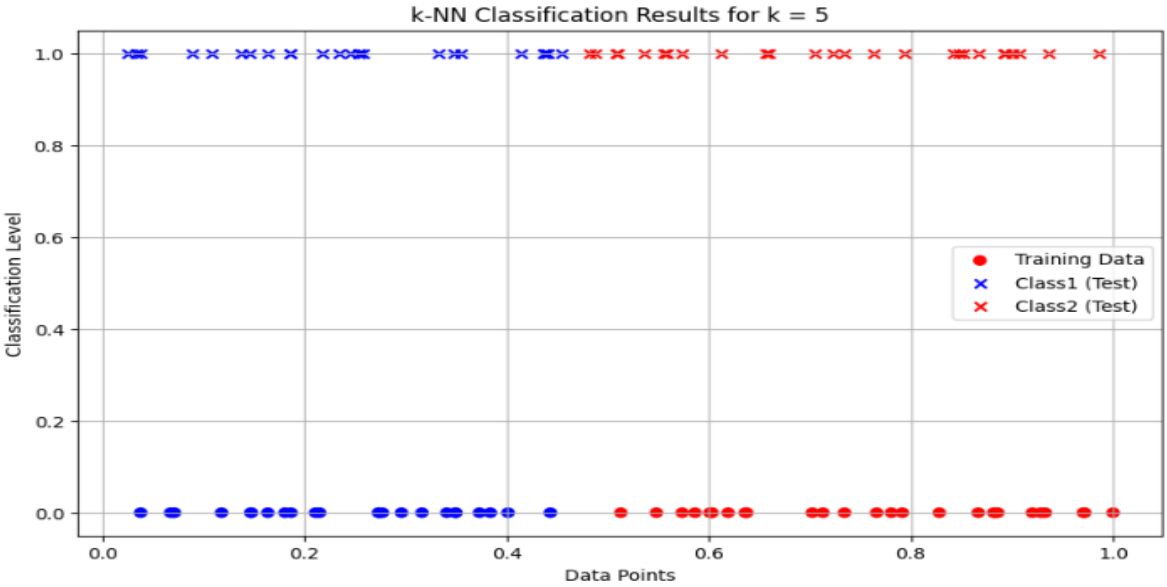

Decision Tree - Breast Cancer Dataset
True
False

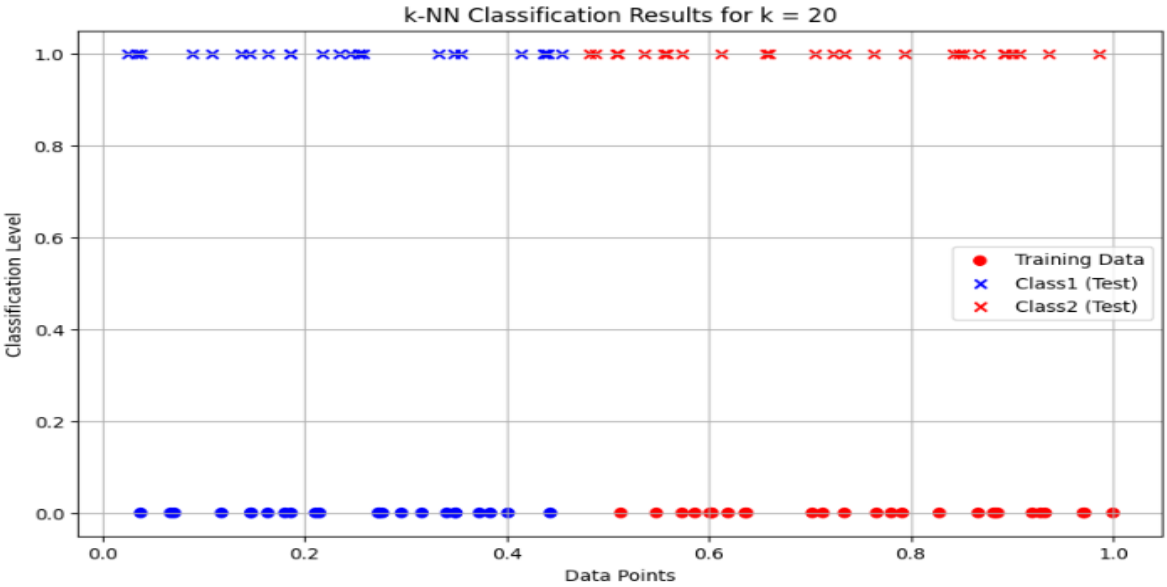

k-NN Classification Results for k = 1
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)

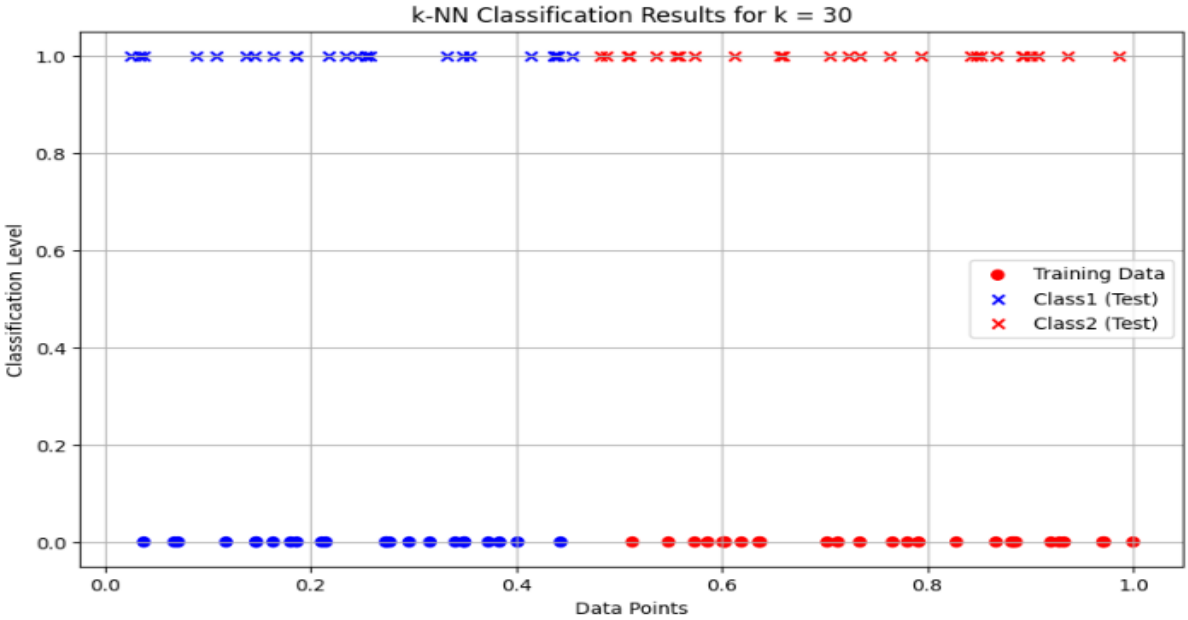

k-NN Classification Results for k = 2
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)


k-NN Classification Results for k = 3
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)


k-NN Classification Results for k = 4
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)


k-NN Classification Results for k = 5
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0


k-NN Classification Results for k = 20
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0


k-NN Classification Results for k = 30
Classification Level
Data Points
Training Data
Class1 (Test)
Class2 (Test)
0.0
0.2
0.4
0.6
0.8
1.0